

analytical_exercises/plots_em.py

```

1 import os
2 import numpy as np
3 import matplotlib.pyplot as plt
4 from mpl_toolkits.mplot3d import Axes3D # needed for 3D plots
5
6 # -----
7 # 0. Ensure 'plots/' directory exists
8 # -----
9 # If 'plots' does not exist, it is created. If it exists, nothing happens.
10 os.makedirs("plots", exist_ok=True)
11
12 # -----
13 # 1. Define x1 and x2 ranges
14 # -----
15
16 # x1 from 0 to 6 (covers our samples 1, 2, 4 and the exponential tail)
17 x1 = np.linspace(0, 6, 200)
18
19 # x2 ranges:
20 #   - Before EM: [0, 3]
21 #   - After EM : [0, 5]
22 x2_before = np.linspace(0, 3, 150)
23 x2_after  = np.linspace(0, 5, 150)
24
25 # Create 2D grids (mesh) of x1 and x2 values
26 X1_before, X2_before = np.meshgrid(x1, x2_before)
27 X1_after,  X2_after  = np.meshgrid(x1, x2_after)
28
29 # -----
30 # 2. Define p(x1, x2) before and after EM
31 # -----
32 # We only care about the *shape* of the distribution (unnormalized):
33 #   Before EM:  $p(x1) \propto \exp(-2 x1)$ ,  $p(x2) = \text{const on } [0, 3]$ 
34 #   After EM :  $p(x1) \propto \exp(-3/7 x1)$ ,  $p(x2) = \text{const on } [0, 5]$ 
35
36 # Before EM
37 p1_before = np.exp(-2 * X1_before) # exponential in x1
38 p2_before = np.ones_like(X2_before) # uniform in x2
39 p_before  = p1_before * p2_before # joint p(x1,x2) up to a constant
40
41 # After EM
42 p1_after = np.exp(-3 * X1_after / 7.0) # slower-decaying exponential
43 p2_after = np.ones_like(X2_after) # uniform on [0,5]
44 p_after  = p1_after * p2_after # joint p(x1,x2) up to a constant
45
46 # -----
47 # 3. Plot the two surfaces side by side
48 # -----
49
50 fig = plt.figure(figsize=(10, 4))
51

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52 # Before EM
53 ax1 = fig.add_subplot(1, 2, 1, projection='3d')
54 ax1.plot_surface(X1_before, X2_before, p_before)
55 ax1.set_title('p(x1,x2) at  $\theta(0)$  of EM')
56 ax1.set_xlabel('x1')
57 ax1.set_ylabel('x2')
58 ax1.set_zlabel('density')
59
60 # After EM
61 ax2 = fig.add_subplot(1, 2, 2, projection='3d')
62 ax2.plot_surface(X1_after, X2_after, p_after)
63 ax2.set_title('p(x1,x2) at  $\theta(1)$  of EM')
64 ax2.set_xlabel('x1')
65 ax2.set_ylabel('x2')
66 ax2.set_zlabel('density')
67
68 plt.tight_layout()
69
70 # -----
71 # 4. Save figure inside 'plots/' directory
72 # -----
73 output_path = "plots/EM_Plots.png"
74 fig.savefig(output_path)
75
```